# Deck 03 · Transforms, forward kinematics, and the Jacobian

## 2.3 Homogeneous transforms and forward kinematics

## 2.4 The Jacobian and differential inverse kinematics

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**Given every encoder reading, where is the hand?**
**Which way does it move when each motor turns a little, and where does that stop working?**

# The change-of-frame equation, and two extensions

$$ {}^{A}\mathbf{p}^{P} = {}^{A}\mathbf{p}^{B} + {}^{A}\mathbf{R}^{B}\,{}^{B}\mathbf{p}^{P} $$

1. **Chains.** Seven frames in a row → one matrix product: the **homogeneous transform**. With joint angles inside: **forward kinematics** $\varphi(\mathbf{q})$.
2. **Derivatives.** Which way the hand moves per joint: the **Jacobian** $\mathbf{J}(\mathbf{q}) = \partial\varphi/\partial\mathbf{q}$.

# Why a $4 \times 4$ matrix

$$ {}^{W}\mathbf{p}^{E} = {}^{W}\mathbf{p}^{1} + {}^{W}\mathbf{R}^{1}\left({}^{1}\mathbf{p}^{2} + {}^{1}\mathbf{R}^{2}\left({}^{2}\mathbf{p}^{3} + \dots\right)\right) $$

Correct, but unreadable, and it grows one nesting level per joint.

**The trick:** append a $1$ to every position and put rotation and translation into one matrix.

# The homogeneous transform

$$
{}^{A}\mathbf{X}^{B} = \begin{bmatrix} {}^{A}\mathbf{R}^{B} & {}^{A}\mathbf{p}^{B} \\ \mathbf{0}^{\top} & 1 \end{bmatrix} \in \mathrm{SE}(3), \qquad {}^{A}\tilde{\mathbf{p}}^{P} = {}^{A}\mathbf{X}^{B}\,{}^{B}\tilde{\mathbf{p}}^{P}
$$

$$
\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}^{\top} & 1 \end{bmatrix} \begin{bmatrix} {}^{B}\mathbf{p}^{P} \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}\,{}^{B}\mathbf{p}^{P} + \mathbf{p} \\ 1 \end{bmatrix}
$$

The first three rows are the change-of-frame equation of 2.2; the fourth says $1 = 1$.

# Composition: inner letters match

$$ {}^{A}\mathbf{X}^{C} = {}^{A}\mathbf{X}^{B}\,{}^{B}\mathbf{X}^{C} $$

Blocks: ${}^{A}\mathbf{R}^{C} = {}^{A}\mathbf{R}^{B}\,{}^{B}\mathbf{R}^{C}$ and ${}^{A}\mathbf{p}^{C} = {}^{A}\mathbf{p}^{B} + {}^{A}\mathbf{R}^{B}\,{}^{B}\mathbf{p}^{C}$.

RAI prints it as $T_{W\to C} = T_{W\to A}T_{A\to B}T_{B\to C}$; we write the monogram form.

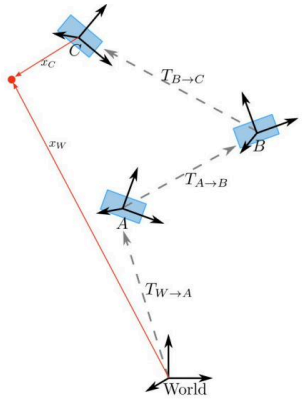
Composition of transforms

*X ~ T : same*

*here:*

$$ {}^{W}\mathrm{X}^{A} \equiv \mathrm{T}_{W\to A} $$

$$ x_W \equiv {}^{W}\mathrm{p}^{\mathrm{red_point}}_{W} $$




$$ \mathrm{T}_{W\to C} \equiv \mathrm{T}_{W\to A}\mathrm{T}_{A\to B}\mathrm{T}_{B\to C} $$

$$ x_W = \mathrm{T}_{W\to A}\mathrm{T}_{A\to B}\mathrm{T}_{B\to C}\,x_C $$

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# Invert a pose: subtract the offset, then rotate back

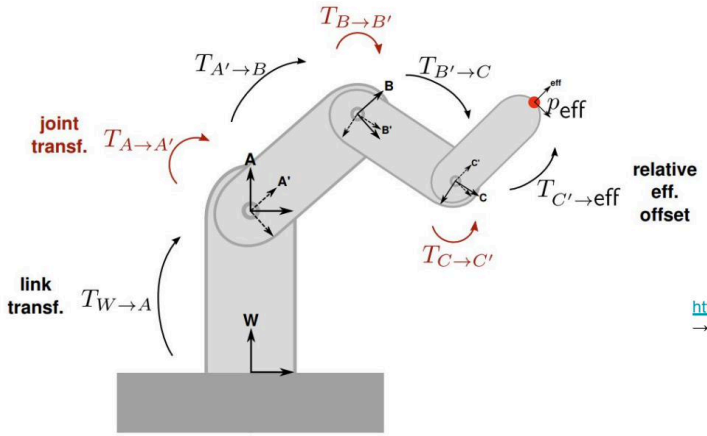
$$
{}^{B}\mathbf{p}^{P} = \mathbf{R}^{\top}({}^{A}\mathbf{p}^{P} - \mathbf{p})
$$

$$
\left({}^{A}\mathbf{X}^{B}\right)^{-1} = {}^{B}\mathbf{X}^{A} = \begin{bmatrix} \mathbf{R}^{\top} & -\mathbf{R}^{\top}\mathbf{p} \\ \mathbf{0}^{\top} & 1 \end{bmatrix}
$$

Transpose the **rotation**; rotate and negate the **translation**.

For a desired pose ${}^{A}\mathbf{X}^{B_d}$, the relative pose error is $\left({}^{A}\mathbf{X}^{B}\right)^{-1}\,{}^{A}\mathbf{X}^{B_d} = {}^{B}\mathbf{X}^{B_d}$: the desired frame seen from the current one, identity when the poses agree.

# A robot is a kinematic tree




- A *kinematic structure* is a graph (usually tree or chain) of rigid **links** and **joints**

https://github.com/MarcToussaint/rai-robotModels
→ check .g files

b

$$T_{W\to\text{eff}}(q) = T_{W\to A}\, T_{A\to A'}(q)\, T_{A'\to B}\, T_{B\to B'}(q)\, T_{B'\to C}\, T_{C\to C'}(q)\, T_{C'\to\text{eff}}$$ 26

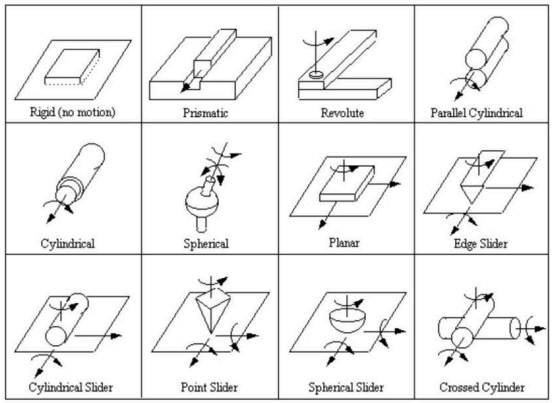
- **Links**: constant transforms, from the geometry
- **Joints**: a hinge or a slider, one number $q_i$ each; revolute turns, prismatic slides
- Walk from the world to the end-effector and multiply, one factor per edge:

$$^{W}\mathbf{X}^{E}(\mathbf{q}) = \mathbf{X}_0\, \mathbf{X}_1(q_1)\, \mathbf{X}_2(q_2) \cdots \mathbf{X}_n(q_n)\, \mathbf{X}_{n+1}$$

**Forward kinematics** $\varphi(\mathbf{q})$: constant factors from the geometry, one moving factor per joint.

# Joint types, and the tree as a `.g` file

Joint types




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our arms use only revolute and prismatic joints; the other types are combinations of the two

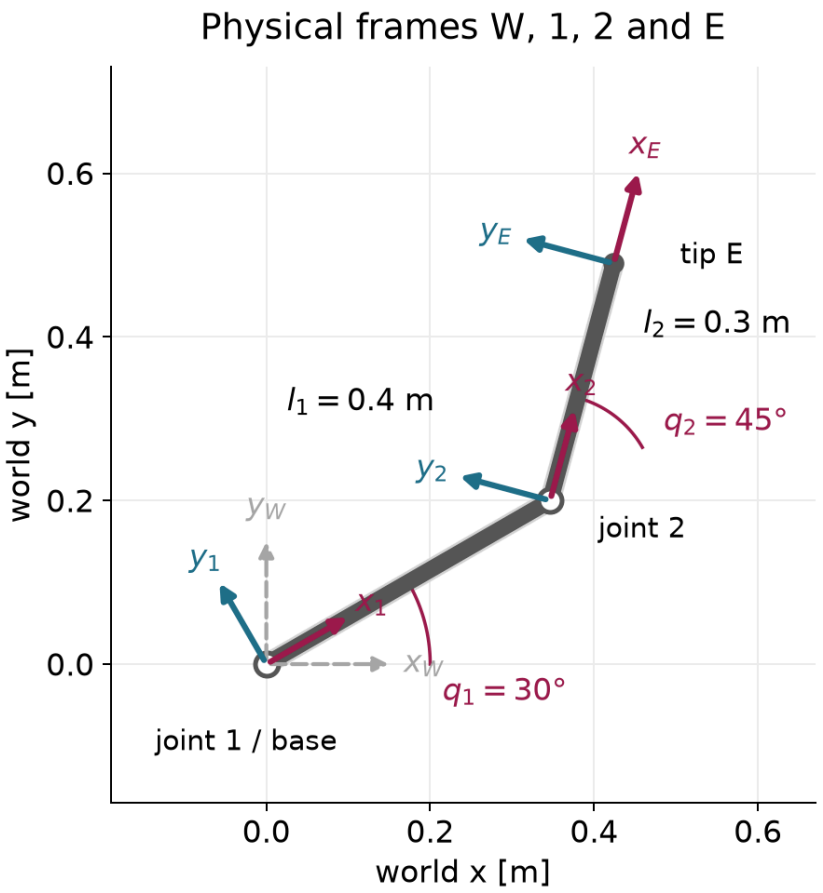
$$\mathbf{X}_{\text{rev}}(q_i) = \begin{bmatrix} \mathbf{R}_z(q_i) & 0 \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad \mathbf{X}_{\text{pri}}(q_i) = \begin{bmatrix} \mathbf{I} & (q_i, 0, 0)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

```
joint1 (world): {
  joint: hingeZ, q: [0.5236]
}
joint2 (joint1): {
  Q: [0.40 0 0],
  joint: hingeZ, q: [0.7854]
}
endeff (joint2): {
  Q: [0.30 0 0], shape: marker
}
```

The tree, written down: parent in parentheses, constant offsets as `Q:`, one number per `joint:`.

code/two_link.g

# The two-link planar arm




$$\mathbf{q} = (30^\circ, 45^\circ), \qquad l_1 = 0.4\,\mathrm{m}, \quad l_2 = 0.3\,\mathrm{m}$$

Frame $1$ sits at the base and points along link 1. Frame $2$ sits at the elbow and points along link 2. Frame $E$ sits at the tip.

The elbow angle is relative to link 1, so the tip's heading is

$$q_1 + q_2 = 75^\circ.$$

We will compute the pose of $E$ in $W$ for **this one fixed configuration**.

# Four factors describe one fixed configuration

The first turn aligns the axes with link 1.

$$\mathbf{R}(30^\circ) \approx \begin{bmatrix} .8660 & -.5000 & 0 \\ .5000 & .8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The second turn aligns the axes with link 2.

$$\mathbf{R}(45^\circ) \approx \begin{bmatrix} .7071 & -.7071 & 0 \\ .7071 & .7071 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The first shift reaches the elbow in those axes.

$$\mathbf{T}(.4, 0) = \begin{bmatrix} 1 & 0 & .4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The second shift reaches the tip in those axes.

$$\mathbf{T}(.3, 0) = \begin{bmatrix} 1 & 0 & .3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$^{W}\mathbf{X}^{E} = \underbrace{\mathbf{R}(q_1)}_{^{W}\mathbf{X}^{1}}\underbrace{\mathbf{T}(l_1, 0)}_{^{1}\mathbf{X}^{2}}\underbrace{\mathbf{R}(q_2)\mathbf{T}(l_2, 0)}_{^{2}\mathbf{X}^{E}}$$

Here $\mathbf{R}(\theta)$ and $\mathbf{T}(x, y)$ are homogeneous $3 \times 3$ factors. The rotation block is $\mathbf{R}_2(\theta)$.

# Appending a factor rotates its shift

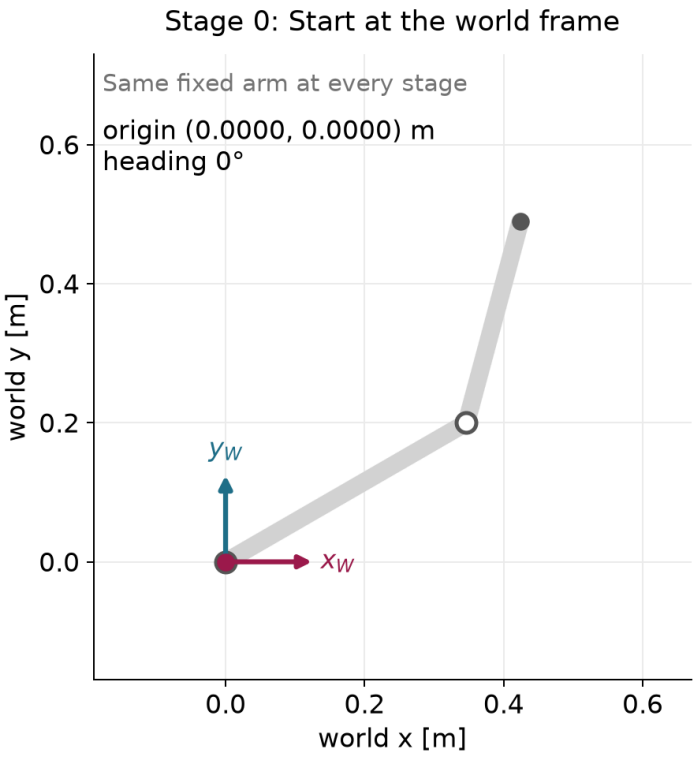
Multiplication updates the axes and the origin separately.

$$\begin{bmatrix} \mathbf{R}_{\text{old}} & \mathbf{p}_{\text{old}} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{\text{next}} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{\text{old}}\mathbf{R}_{\text{next}} & \mathbf{p}_{\text{old}} + \mathbf{R}_{\text{old}}\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$$

The next shift is in the **current frame's axes**, so rotate it before adding it to the world position.

$$\mathbf{X}^{(0)} = \mathbf{I}_3, \qquad \mathbf{X}^{(s+1)} = \mathbf{X}^{(s)}\,(\text{next factor})$$

The superscript $s$ counts construction stages. It is neither a link number nor time.



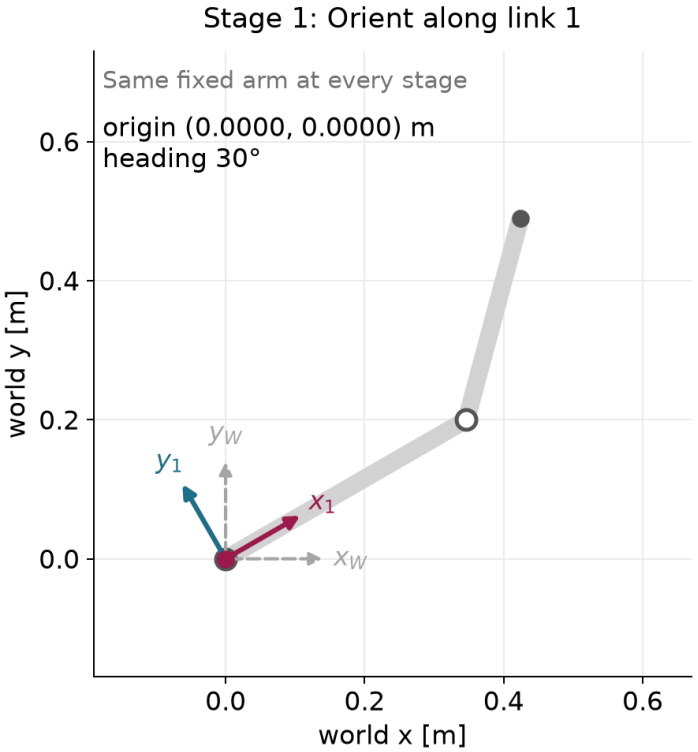
# Turn the frame at joint 1

The first joint sets the direction of link 1. Its origin stays at the base.

$$\mathbf{X}^{(1)} = {}^{W}\mathbf{X}^{1} = \mathbf{I}_3\mathbf{R}(30^\circ) \approx \begin{bmatrix} .8660 & -.5000 & 0 \\ .5000 & .8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{p}_{\text{new}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \mathbf{I}_2 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The first column points at $30^\circ$.



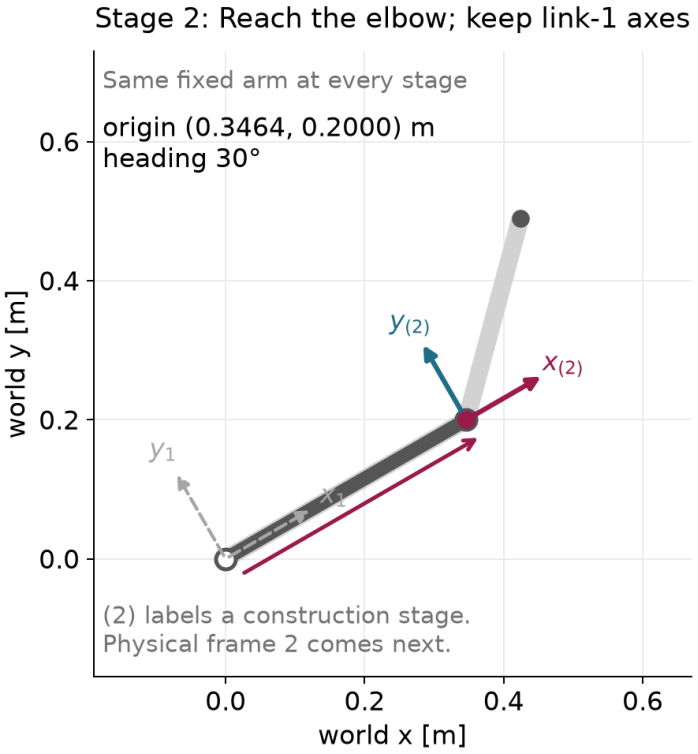
# Shift to joint 2 along the turned x-axis

The elbow is $0.4$ m along link 1, so its displacement must be rotated into world axes.

$$\mathbf{p}_{\text{new}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ \\ \sin 30^\circ & \cos 30^\circ \end{bmatrix} \begin{bmatrix} .4 \\ 0 \end{bmatrix} \approx \begin{bmatrix} .3464 \\ .2000 \end{bmatrix}$$

$$\mathbf{X}^{(2)} = \mathbf{X}^{(1)}\mathbf{T}(.4, 0) \approx \begin{bmatrix} .8660 & -.5000 & .3464 \\ .5000 & .8660 & .2000 \\ 0 & 0 & 1 \end{bmatrix}$$

The axes still point at $30^\circ$. This intermediate construction frame is at the elbow; physical frame $2$ also includes the next turn.



# Turn the frame at joint 2

The elbow turn adds $45^\circ$ to the heading. Its zero translation leaves the elbow origin fixed.

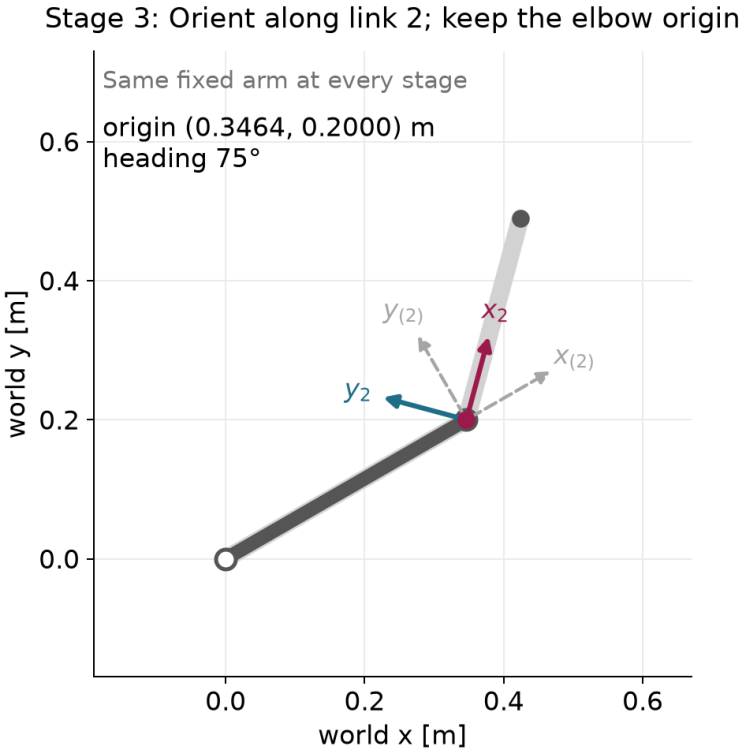
$$\mathbf{X}^{(3)} = \mathbf{X}^{(2)}\mathbf{R}(45^\circ) = {}^{W}\mathbf{X}^{2} \approx \begin{bmatrix} .2588 & -.9659 & .3464 \\ .9659 & .2588 & .2000 \\ 0 & 0 & 1 \end{bmatrix}$$

Each new rotation entry is a row times a column. For the new x-axis:

$$\cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ \approx .2588,$$

$$\sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ \approx .9659.$$

These are the world components of the $75^\circ$ x-axis.



# Shift to the tip along its x-axis

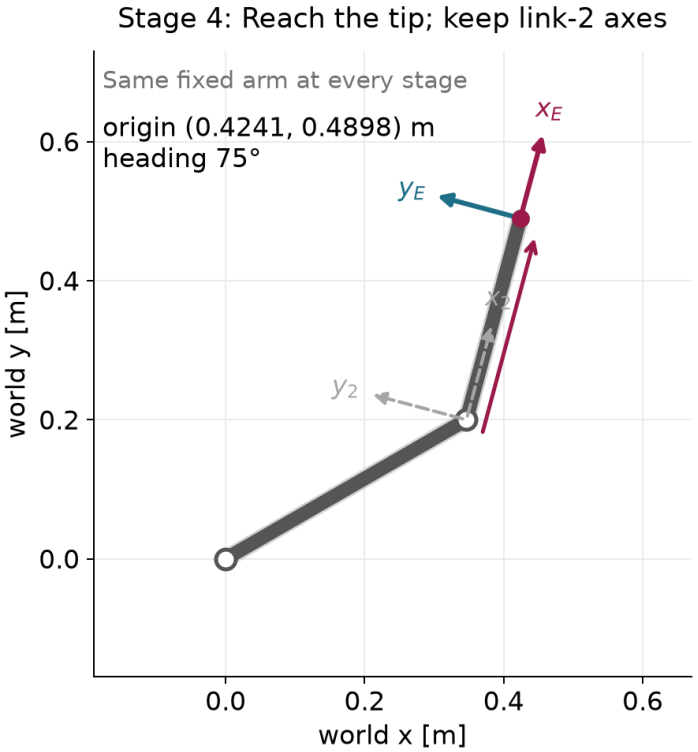
The tip is $0.3$ m along the new x-axis. Add that displacement to the elbow position.

$$\mathbf{p}_{\text{new}} = \begin{bmatrix} .4\cos 30^\circ \\ .4\sin 30^\circ \end{bmatrix} + \mathbf{R}_2(75^\circ) \begin{bmatrix} .3 \\ 0 \end{bmatrix}$$

$$\mathbf{p}_{\text{new}} \approx \begin{bmatrix} .34641 + .07765 \\ .20000 + .28978 \end{bmatrix} \approx \begin{bmatrix} .4241 \\ .4898 \end{bmatrix}$$

$$\mathbf{X}^{(4)} = \mathbf{X}^{(3)}\mathbf{T}(.3, 0) = {}^{W}\mathbf{X}^{E} \approx \begin{bmatrix} .2588 & -.9659 & .4241 \\ .9659 & .2588 & .4898 \\ 0 & 0 & 1 \end{bmatrix}$$

The translation preserves the $75^\circ$ heading.



# Multiply the full pose chain in one shot

The four stages are one matrix product. The displayed entries are rounded.

$$
{}^{W}\mathbf{X}^{E} \approx \underbrace{\begin{bmatrix} .8660 & -.5000 & 0 \\ .5000 & .8660 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{R}(30^\circ)} \underbrace{\begin{bmatrix} 1 & 0 & .4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}(.4,0)} \underbrace{\begin{bmatrix} .7071 & -.7071 & 0 \\ .7071 & .7071 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{R}(45^\circ)} \underbrace{\begin{bmatrix} 1 & 0 & .3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{T}(.3,0)}
$$

$$
\approx \begin{bmatrix} .2588 & -.9659 & .4241 \\ .9659 & .2588 & .4898 \\ 0 & 0 & 1 \end{bmatrix}.
$$

The coloured columns give the tip's axes in $W$. Its position column, in metres, gives the task we will differentiate:

$$
\varphi(\mathbf{q}) = {}^{W}\mathbf{p}^{E}(\mathbf{q}) = \begin{bmatrix} l_1 \cos q_1 + l_2 \cos(q_1 + q_2) \\ l_1 \sin q_1 + l_2 \sin(q_1 + q_2) \end{bmatrix}.
$$

code/fk_chain_walkthrough.py · code/two_link.py · Drake's two-link check agrees

# Apply the same product to coordinates from the right

The tip's origin has coordinates zero in $E$. Each factor rewrites the **same point** in the preceding axes.

$$ {}^{W}\tilde{\mathbf{p}}^{E} = \mathbf{R}(30^\circ)\mathbf{T}(.4, 0)\mathbf{R}(45^\circ)\mathbf{T}(.3, 0) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} $$

| Apply this factor next | The point's homogeneous coordinates | These coordinates use |
|---|---|---|
| Start at the right | $(0, 0, 1)^\top$ | tip frame $E$ |
| $\mathbf{T}(.3, 0)$ | $(.3, 0, 1)^\top$ | physical frame $2$ |
| $\mathbf{R}(45^\circ)$ | $\approx (.2121, .2121, 1)^\top$ | stage-2 elbow axes |
| $\mathbf{T}(.4, 0)$ | $\approx (.6121, .2121, 1)^\top$ | physical frame $1$ |
| $\mathbf{R}(30^\circ)$ | $\approx (.4241, .4898, 1)^\top$ | world frame $W$ |

# Lift the planar pose into 3D

We insert a physical z-coordinate and keep the homogeneous $1$ last.

$$\underbrace{\begin{bmatrix} .2588 & -.9659 & .4241 \\ .9659 & .2588 & .4898 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{planar pose, rounded}} \longrightarrow \underbrace{\begin{bmatrix} .2588 & -.9659 & 0 & .4241 \\ .9659 & .2588 & 0 & .4898 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{same pose in 3D, rounded}}$$

$$(x, y, 1)^\top \longrightarrow (x, y, 0, 1)^\top$$

The third rotation column is the unchanged z-axis. The third position entry is zero. A spatial arm uses these same $4 \times 4$ products with axes that point in different directions.

# The FR3 uses the same transform chain

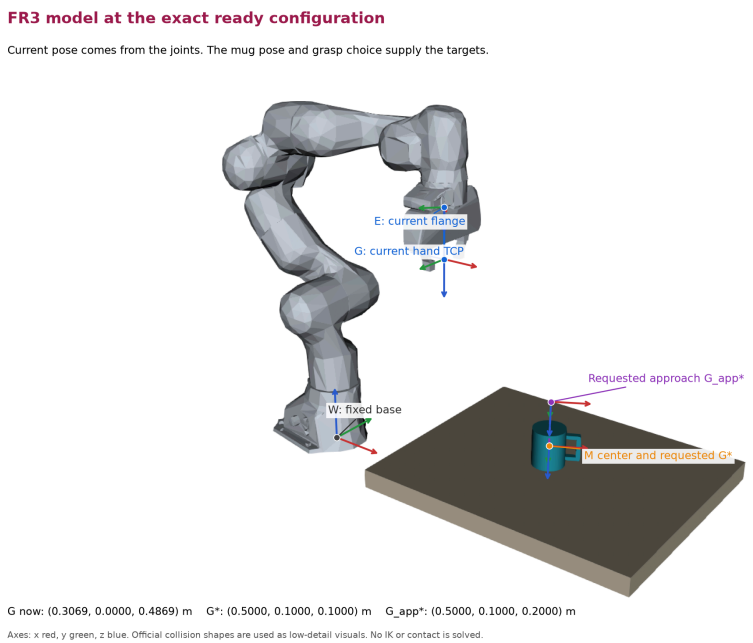
The model supplies each fixed link offset and joint axis. Joint readings supply the seven angles. We use this supplied configuration.

$$\mathbf{q} = (0, -\pi/4, 0, -3\pi/4, 0, \pi/2, \pi/4).$$

Each joint combines its fixed placement with a turn about its local z-axis:

$$^{i-1}\mathbf{X}^{i}(q_i) = \begin{bmatrix} \mathbf{R}_{\text{origin},i} & \mathbf{p}_{\text{origin},i} \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_z(q_i) & \mathbf{0} \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

Here the physical FR3 links are numbered $0$ through $7$. The fixed base is $W$; $E$ is the flange; $G$ is the hand's tool center point.




The FR3 scene uses the manufacturer's frame chain. The mug pose is supplied.

franka_description 2.9.0 · drake/01c-fr3-grasp-chain ▶ run it live

# Build one FR3 joint transform from the model

Joint $2$ has zero fixed translation. Its fixed rotation is about x, and its variable rotation is about local z.

$$
{}^{1}\mathbf{X}^{2}(-\pi/4) = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{fixed } \mathbf{R}_x(-\pi/2)} \underbrace{\begin{bmatrix} c & c & 0 & 0 \\ -c & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{joint } \mathbf{R}_z(-\pi/4)} = \begin{bmatrix} c & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ c & -c & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad c = \frac{\sqrt{2}}{2}.
$$

Joint $1$ has zero angle, so its pose is a pure $0.333$ m rise from the base.

$$
{}^{W}\mathbf{X}^{2} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & .333 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{{}^{W}\mathbf{X}^{1}} \underbrace{\begin{bmatrix} c & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ c & -c & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{{}^{1}\mathbf{X}^{2}} = \begin{bmatrix} c & c & 0 & 0 \\ 0 & 0 & 1 & 0 \\ c & -c & 0 & .333 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

# Carry a link offset into world axes

Link $3$ lies $0.316$ m along link $2$'s negative y-axis. Rotate this local offset:

$$ {}^{W}\mathbf{p}^{3} = {}^{W}\mathbf{p}^{2} + {}^{W}\mathbf{R}^{2} \begin{bmatrix} 0 \\ -.316 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ .333 \end{bmatrix} + \begin{bmatrix} -.316/\sqrt{2} \\ 0 \\ .316/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} -.2234 \\ 0 \\ .5564 \end{bmatrix} \ \mathrm{m}. $$

Repeat for all seven joints, then append the fixed flange transform:

$$ {}^{W}\mathbf{X}^{E}(\mathbf{q}) = {}^{W}\mathbf{X}^{0} \left( \prod_{i=1}^{7} {}^{i-1}\mathbf{X}^{i}(q_i) \right) {}^{7}\mathbf{X}^{E} \approx \begin{bmatrix} .7071 & -.7071 & 0 & .3069 \\ -.7071 & -.7071 & 0 & 0 \\ 0 & 0 & -1 & .5903 \\ 0 & 0 & 0 & 1 \end{bmatrix}. $$

notes §5(m.1) lists every fixed factor · facts.json stores every intermediate product

# From the flange to the grasp point

The manufacturer places the hand's tool center point $G$ between the fingers. Its offset from $E$ is fixed.

$$
{}^{E}\mathbf{X}^{G} \approx \begin{bmatrix} .7071 & .7071 & 0 & 0 \\ -.7071 & .7071 & 0 & 0 \\ 0 & 0 & 1 & .1034 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

This rotates the hand axes by $-45^\circ$ about the flange z-axis and shifts $0.1034$ m along that axis.

$$
{}^{W}\mathbf{X}^{G}(\mathbf{q}) = {}^{W}\mathbf{X}^{E}(\mathbf{q})\,{}^{E}\mathbf{X}^{G} \approx \begin{bmatrix} 1 & 0 & 0 & .3069 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & .4869 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
\mathbf{q} = (0, -\pi/4, 0, -3\pi/4, 0, \pi/2, \pi/4).
$$

$$
{}^{W}\mathbf{p}^{G} = {}^{W}\mathbf{p}^{E} + {}^{W}\mathbf{R}^{E} \begin{bmatrix} 0 \\ 0 \\ .1034 \end{bmatrix} \approx \begin{bmatrix} .3069 \\ 0 \\ .5903 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -.1034 \end{bmatrix} = \begin{bmatrix} .3069 \\ 0 \\ .4869 \end{bmatrix} \ \text{m}.
$$

# From the mug to the desired grasp

The supplied mug frame $M$ is at its center, with a $30^\circ$ yaw.

$$ {}^{W}\mathbf{X}^{M} \approx \begin{bmatrix} .8660 & -.5000 & 0 & .50 \\ .5000 & .8660 & 0 & .10 \\ 0 & 0 & 1 & .10 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Choose the desired tool frame $G^*$ at that center, with its x-axis along the mug handle and its z-axis down.

$$ {}^{M}\mathbf{X}^{G^*} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Matching inner letters gives the requested pose in world coordinates.

$$ {}^{W}\mathbf{X}^{G^*} = {}^{W}\mathbf{X}^{M}{}^{M}\mathbf{X}^{G^*} \approx \begin{bmatrix} .8660 & .5000 & 0 & .50 \\ .5000 & -.8660 & 0 & .10 \\ 0 & 0 & -1 & .10 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

The chosen grasp flips the mug's y- and z-axes. Its zero offset keeps the mug-center position.

**Current $G$ comes from the joints. Desired $G^*$ comes from the mug and the grasp choice.**

# Choose an approach pose before asking for joints

**1. Choose the offset.** Stand off by $0.10$ m along negative grasp z. That axis points down, so the approach is above the mug.

**2. Compose the poses.** Append the local translation to the desired grasp pose.

$$\underbrace{\begin{bmatrix} .8660 & .5000 & 0 & .50 \\ .5000 & -.8660 & 0 & .10 \\ 0 & 0 & -1 & .10 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{{}^{W}\mathbf{X}^{G^*}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -.10 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{{}^{G^*}\mathbf{X}^{G^*_{\text{app}}}} \approx \underbrace{\begin{bmatrix} .8660 & .5000 & 0 & .50 \\ .5000 & -.8660 & 0 & .10 \\ 0 & 0 & -1 & .20 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{{}^{W}\mathbf{X}^{G^*_{\text{app}}}}$$

**3. Ask for the joints.** Inverse kinematics (IK) seeks joint values that put the actual tool at this approach pose.

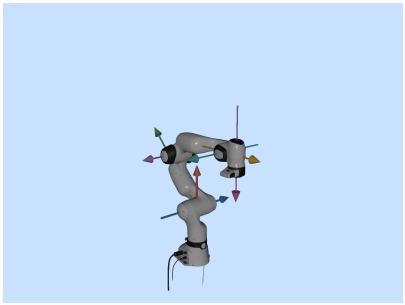
$${}^{W}\mathbf{X}^{G}(\mathbf{q}) = {}^{W}\mathbf{X}^{G^*_{\text{app}}}.$$

# Live demo: the Panda's 7-joint arm in Drake

`fk_jacobian_demo.py --meshcat`

- Seven joint sliders; the flange follows: **forward kinematics, live**. The coloured arrows show the positive joint axes
- Predict: joint 1 turns in the positive direction from the ready pose. Which way does the hand go?
- Watch ${}^{W}\mathbf{p}^{E}$ change; the console's rank and $\sigma_{\min}$ come back on the Panda singularity slide

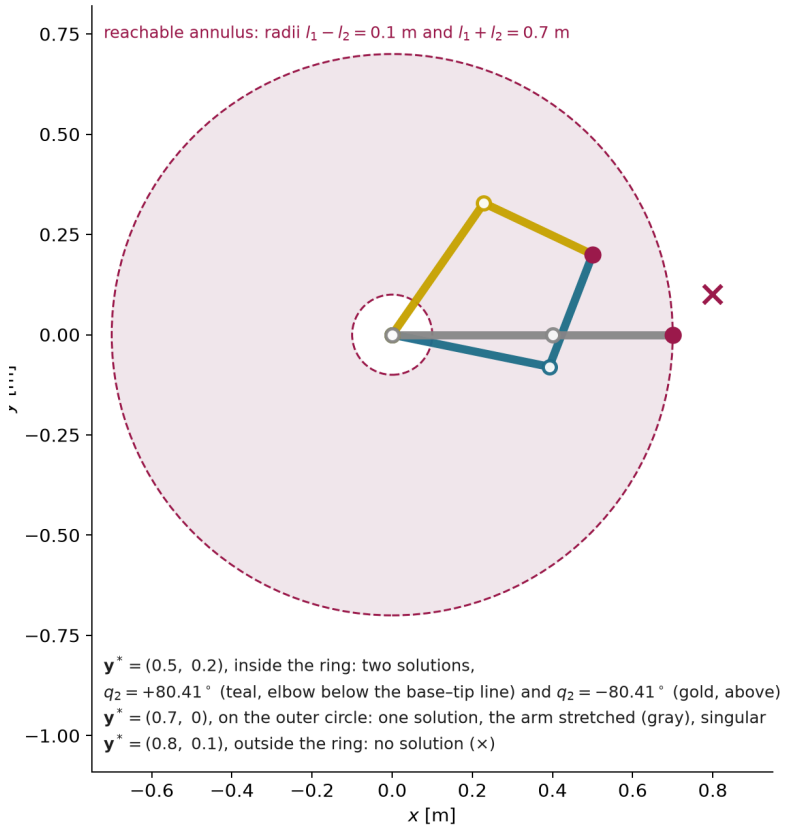
**In the stack:** item 1 of the map. The **Goal box** asks it to reach a target; **Reference motion** uses it for collisions.



ready pose, positive joint-axis arrows (--png)

drake/01-fk-jacobian ▶ run it live

# Inverse kinematics: from a target to joint positions




Given a desired tip position or pose, find joint positions. This is **inverse kinematics (IK)**.

For the two-link position task, seek

$$\varphi(\mathbf{q}) = \mathbf{y}^*.$$

- Two-link position task: **none** outside the ring; **one** on a boundary; **two** inside
- Seven joints, full pose fixed: often a **continuum** of postures

Our two-link arm admits an **analytic formula**.

We focus on **numerical IK**: one formulation for different arms, posture preferences and constraints.

**Deck 04:** derive error-reducing steps using FK and its Jacobian.
**Deck 05:** add joint limits and clearance constraints.

# Summary of 2.3

- A pose is one $4 \times 4$ matrix; the extra $1$ turns "rotate and add" into one product
- Compose by multiplying with matching inner letters; invert with $\mathbf{R}^\top$ and $-\mathbf{R}^\top \mathbf{p}$
- The chain product gives the tip pose. A position-only task keeps its position coordinates.

| Question | Given | Compute or seek |
|---|---|---|
| Forward kinematics | Joint positions | Tip position or pose |
| Inverse kinematics | Desired tip position or pose | Joint positions |

Our two-link FK example used joint angles $(30^\circ, 45^\circ)$ to compute tip position $(0.4241, 0.4898)$ m and heading $75^\circ$.

Next, we predict **tip velocity from joint velocities**. That local motion map will also help us improve an IK guess.

# Break

Second half: which way does the tip move when a joint turns, and where does that fail?

# The Jacobian: a table of small joint effects

Forward kinematics tells us where the tip is. To move it, we also need to predict **how joint velocities make the tip move**.

$$\Delta \mathbf{x} \approx \mathbf{J}(\mathbf{q})\Delta \mathbf{q}, \qquad J_{ij} = \frac{\partial x_i}{\partial q_j}$$

- **Column $j$:** what moving joint $j$ does to the task
- **Row $i$:** how all joints contribute to task component $i$
- $m$ task components, $n$ joints: $\mathbf{J} \in \mathbb{R}^{m \times n}$

Divide by a short time interval and take its limit. This gives **differential forward kinematics**:

$$\underbrace{\dot{\mathbf{x}}}_{\text{task velocity}} = \mathbf{J}(\mathbf{q}) \underbrace{\dot{\mathbf{q}}}_{\text{joint velocities}}$$

$\mathbf{x}$ is the **task-coordinate vector**, such as the tip's position coordinates. Full pose motion uses the six-row geometric Jacobian introduced on the 3D slide.

# The two-link Jacobian, derived

**1. Name each entry.** Rows are tip coordinates; columns are joints.

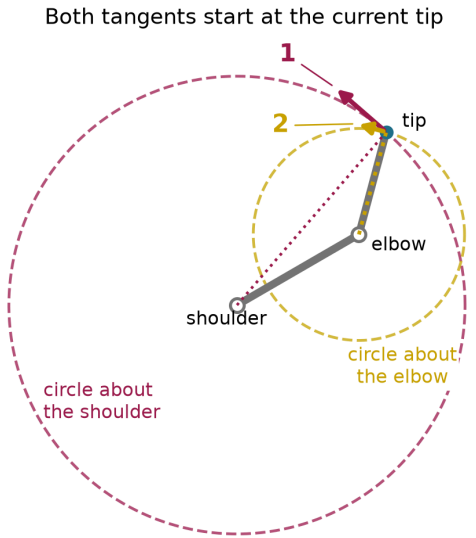
$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \dfrac{\partial x}{\partial q_1} & \dfrac{\partial x}{\partial q_2} \\ \dfrac{\partial y}{\partial q_1} & \dfrac{\partial y}{\partial q_2} \end{bmatrix}$$

**2. Differentiate the position.**

$$x = l_1 c_1 + l_2 c_{12},$$
$$y = l_1 s_1 + l_2 s_{12}.$$

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix}$$

$$c_1 = \cos q_1, \qquad s_1 = \sin q_1,$$
$$c_{12} = \cos(q_1 + q_2), \quad s_{12} = \sin(q_1 + q_2).$$




- **Joint 1 alone:** the tip moves tangent to the shoulder's circle.
- **Joint 2 alone:** the tip moves tangent to the elbow's circle.

Each column gives tip motion **per radian** of that joint.

# Read the numbers as joint effects

**1. Substitute the worked pose and link lengths.**

$$l_1 = 0.4\ \text{m}, \quad l_2 = 0.3\ \text{m}, \quad q_1 = 30^\circ, \quad q_1 + q_2 = 75^\circ.$$

$$\mathbf{J}_{\text{pos}} = \begin{bmatrix} -0.4\sin 30^\circ - 0.3\sin 75^\circ & -0.3\sin 75^\circ \\ 0.4\cos 30^\circ + 0.3\cos 75^\circ & 0.3\cos 75^\circ \end{bmatrix}$$

**2. Evaluate the entries.**

$$\mathbf{J}_{\text{pos}} \approx \begin{bmatrix} -0.4898 & -0.2898 \\ 0.4241 & 0.0776 \end{bmatrix}\ \text{m/rad}$$

**3. Use the same matrix for displacement or velocity.**

**Joint displacement input (rad):**

$$\Delta\mathbf{p} \approx \mathbf{J}_{\text{pos}} \begin{bmatrix} 0.01 \\ 0 \end{bmatrix} \approx \begin{bmatrix} -0.004898 \\ 0.004241 \end{bmatrix}\ \text{m}$$

About **4.90 mm left, 4.24 mm up**.

**Joint velocity input (rad/s):**

$$\dot{\mathbf{p}} = \mathbf{J}_{\text{pos}} \begin{bmatrix} 0.01 \\ 0 \end{bmatrix} \approx \begin{bmatrix} -0.004898 \\ 0.004241 \end{bmatrix}\ \text{m/s}$$

**Differential FK:** about 4.90 mm/s left, 4.24 mm/s up.

code/two_link.py · chapter-01.md runnable example

# First order only

| turn joint 1 by | actual tip motion | $\mathbf{J}\,\Delta\mathbf{q}$ | error |
|---|---|---|---|
| $5^\circ$ | $(-0.0443,\ 0.0351)$ m | $(-0.0427,\ 0.0370)$ m | 2.5 mm |
| $30^\circ$ | $(-0.3017,\ 0.1464)$ m | $(-0.2564,\ 0.2220)$ m | 9 cm |

The Jacobian is a **local** statement about $\varphi$ at one $\mathbf{q}$.

# Add heading: the same arm, one more task row

**1. Describe the planar pose.** Add the gripper's heading to its position coordinates.

$$(\theta, x, y)^\top, \qquad \theta = q_1 + q_2.$$

These are one angle and two position coordinates, not a point in 3D.

**2. Differentiate the heading.**

$$\frac{\partial \theta}{\partial q_1} = 1, \qquad \frac{\partial \theta}{\partial q_2} = 1.$$

$$\begin{aligned} \dot{\theta} &= \frac{\partial \theta}{\partial q_1}\dot{q}_1 + \frac{\partial \theta}{\partial q_2}\dot{q}_2 \\ &= \dot{q}_1 + \dot{q}_2. \end{aligned}$$

**3. Add the heading row above the position rows.**

$$\mathbf{J}_{\text{full}} = \begin{bmatrix} \frac{\partial \theta}{\partial q_1} & \frac{\partial \theta}{\partial q_2} \\ \frac{\partial x}{\partial q_1} & \frac{\partial x}{\partial q_2} \\ \frac{\partial y}{\partial q_1} & \frac{\partial y}{\partial q_2} \end{bmatrix}$$

**4. Compute the gripper motion at this pose.**

$$\begin{bmatrix} \dot{\theta} \\ \dot{x} \\ \dot{y} \end{bmatrix} \approx \begin{bmatrix} 1 & 1 \\ -0.4898 & -0.2898 \\ 0.4241 & 0.0776 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}.$$

The inputs $\dot{q}_1$ and $\dot{q}_2$ are **joint angular velocities** (rad/s). The outputs are gripper angular velocity (rad/s) and tip velocity (m/s).

# In 3D: three turning rates, three moving rates

**Reaching the right point is not enough for a grasp. The fingers must also face the right way. Angular velocity describes how we turn them toward that orientation.**

$\boldsymbol{\omega}$ is the **angular-velocity vector**, expressed in world axes.

- Its **direction** gives the rotation axis (right-hand rule).
- Its **magnitude** gives angular speed (rad/s).

For the planar arm, rotation is about the world z-axis:

$$\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta} \end{bmatrix}.$$

A zero vector means no instantaneous rotation.

$\mathbf{v}$ is the **linear velocity of $E$'s origin**, in m/s (the flange in the Panda demos).

**Differential forward kinematics:** at the current joint positions, multiply the full Jacobian by joint angular velocities to compute the flange's turning and moving velocities.

**Spatial velocity** (world axes, angular first):

$$^{W}\mathbf{V}^{E} = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_{\text{ang}} \\ \mathbf{J}_{\text{pos}} \end{bmatrix} \dot{\mathbf{q}}$$

| Task | Panda Jacobian |
|---|---|
| Position only: $v_x, v_y, v_z$ | $3 \times 7$ |
| Full motion: $\omega_x, \omega_y, \omega_z, v_x, v_y, v_z$ | $6 \times 7$ |

# Build the angular Jacobian from the joint axes

We reuse the **FR3 ready pose**. Each hinge rotates about its own **local z-axis**.

**1. Read both rotations at the ready pose.** Joint 1 has zero angle and no fixed rotation.

$$ {}^{W}\mathbf{R}^{1} = \mathbf{I}_3, \qquad {}^{W}\mathbf{R}^{2} = \begin{bmatrix} c & c & 0 \\ 0 & 0 & 1 \\ c & -c & 0 \end{bmatrix}. $$

Here $\mathbf{I}_3$ is the identity rotation, and $c = \sqrt{2}/2$.

**2. Convert each unit joint axis to world coordinates.**

$$ {}^{W}\hat{\mathbf{z}}_1 = {}^{W}\mathbf{R}^{1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, $$

$$ {}^{W}\hat{\mathbf{z}}_2 = {}^{W}\mathbf{R}^{2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. $$

**3. Stack the directions and multiply.**

The first two columns of $\mathbf{J}_{\text{ang}}$ are

$$ \begin{bmatrix} {}^{W}\hat{\mathbf{z}}_1 & {}^{W}\hat{\mathbf{z}}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}. $$

Move joints 1 and 2 at **0.1 and 0.2 rad/s**. Hold the others still.

$$ \boldsymbol{\omega} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.2 \\ 0.1 \end{bmatrix} \ \text{rad/s}. $$

Multiply each column by its joint velocity, then add.

FR3 ready pose · drake/01c-fr3-grasp-chain/fr3_grasp_chain.py · essentials §6

# Differential inverse kinematics, the square case

We computed gripper motion from joint velocities. **Now choose the gripper motion and find the joint velocities.** We use the **two-link position task** for the calculation.

To lift a mug, request an **upward tip velocity**:

$$\dot{\mathbf{p}}_{\text{cmd}} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} \ \text{m/s}.$$

The current joint positions determine $\mathbf{J}_{\text{pos}}$.
Solve for the unknown joint velocities:

$$\mathbf{J}_{\text{pos}}(\mathbf{q}) \underbrace{\dot{\mathbf{q}}}_{\text{find}} = \underbrace{\dot{\mathbf{p}}_{\text{cmd}}}_{\text{requested}} .$$

This is **differential inverse kinematics**. When the Jacobian is square and invertible,

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{pos}}^{-1} \dot{\mathbf{p}}_{\text{cmd}}.$$

At the worked joint positions,

$$\mathbf{q} = \begin{bmatrix} 30^\circ \\ 45^\circ \end{bmatrix}, \qquad \mathbf{J}_{\text{pos}} \approx \begin{bmatrix} -0.4898 & -0.2898 \\ 0.4241 & 0.0776 \end{bmatrix} \ \text{m/rad}.$$

The solution is

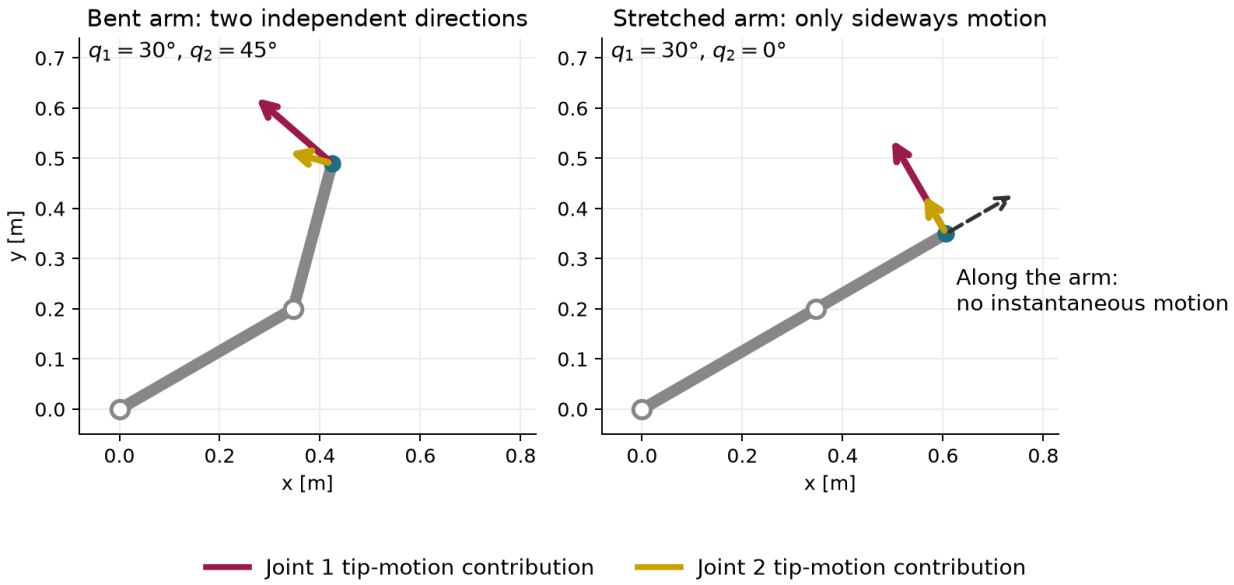
$$\dot{\mathbf{q}} \approx \begin{bmatrix} 0.3415 \\ -0.5772 \end{bmatrix} \ \text{rad/s}.$$

Check the achieved tip velocity:

$$\mathbf{J}_{\text{pos}} \dot{\mathbf{q}} \approx \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} \ \text{m/s}.$$

At these joint positions, specifying both tip-velocity components determines a **unique pair of joint angular velocities**.

# Why the inverse can fail




Open the two-link module. Compare **Worked example** with **Stretched**.

At the stretched pose, both joint effects point **sideways**. No combination produces instantaneous tip motion **along the arm**. Sideways motion remains possible.

**Rank** counts independent tip-motion directions:

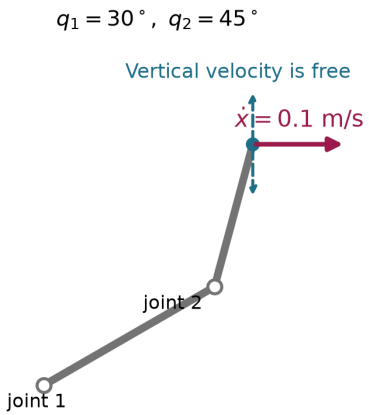
$$\operatorname{rank}(\mathbf{J}_{\text{pos}}): \quad 2 \longrightarrow 1.$$

A drop below the task's maximum attainable rank is a **singularity**.

Near this pose, some tip requests need **large joint velocities**. Optimization I introduces damping to reduce them.

# Several joint motions can satisfy one request

Previously, we requested both tip-velocity components. Now request **only horizontal velocity**; vertical velocity is free.




$$\mathbf{J}_{\text{pos}} \approx \begin{bmatrix} -0.4898 & -0.2898 \\ 0.4241 & 0.0776 \end{bmatrix} \text{ m/rad.}$$

Its first row is $\mathbf{J}_x$, used on the right.

Keep $q_1 = 30^\circ$ and $q_2 = 45^\circ$. Use the horizontal row:

$$\underbrace{\begin{bmatrix} -0.4898 & -0.2898 \end{bmatrix}}_{\mathbf{J}_x} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = 0.1 \text{ m/s}.$$

Both choices satisfy the request:

| Choice | $\dot{q}_1$ (rad/s) | $\dot{q}_2$ (rad/s) | $\dot{x}$ (m/s) |
|---|---|---|---|
| A: joint 1 only | $-0.2042$ | $0$ | $0.1000$ |
| B: joint 2 only | $0$ | $-0.3451$ | $0.1000$ |

**One requested component, two joint velocities:** there are more answers to choose from.

code/two_link.py · code/make_figures.py · essentials §7

# The pseudo-inverse selects one of those answers

Given a requested task velocity, the **pseudo-inverse** is the matrix that selects the joint velocities with the closest achievable match and the smallest norm among ties.

For a wide Jacobian with independent rows (full row rank),

$$\mathbf{J}^{+} = \mathbf{J}^{\top}(\mathbf{J}\mathbf{J}^{\top})^{-1}, \qquad \dot{\mathbf{q}} = \mathbf{J}^{+}\dot{\mathbf{x}}.$$

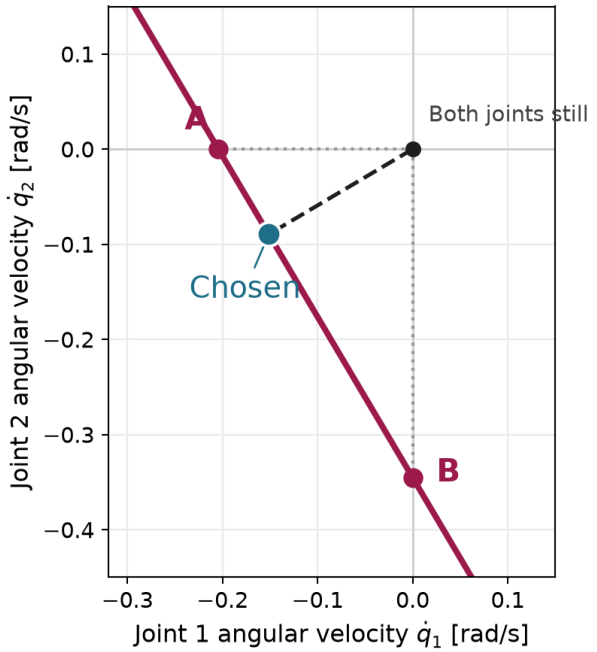
**For this nonzero one-row Jacobian $\mathbf{J}\mathbf{J}^{\top}$** is the scalar sum of squared entries, so transpose the row and divide:

$$\mathbf{J}_x^{+} = \frac{\begin{bmatrix} -0.4898 \\ -0.2898 \end{bmatrix}}{(-0.4898)^2 + (-0.2898)^2} \approx \begin{bmatrix} -1.5123 \\ -0.8948 \end{bmatrix} \ \mathrm{rad/m}.$$

Multiply by the requested horizontal speed:

$$\dot{\mathbf{q}} = \mathbf{J}_x^{+}\,(0.1) \approx \begin{bmatrix} -0.1512 \\ -0.0895 \end{bmatrix} \ \mathrm{rad/s}.$$

This is the valid point **nearest the origin** in the graph.




One equation in two unknowns leaves a one-dimensional set; the equation is linear in $\dot{\mathbf{q}}$, so the set is a line, and moving along it changes only the vertical tip speed.

code/two_link.py · np.linalg.pinv(J_x) · essentials §7

# Extra joint motion can preserve the task

Start from A. Add coordinated joint motion to obtain B:

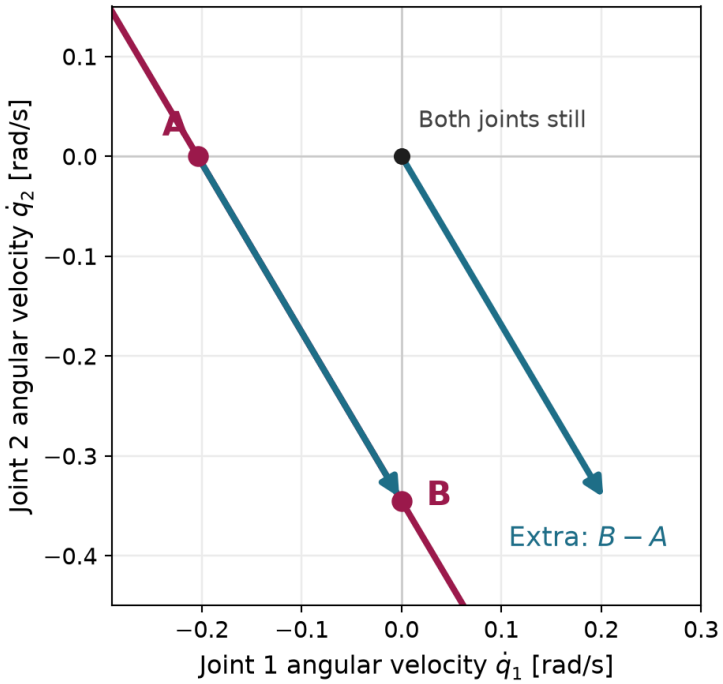
$$\underbrace{\begin{bmatrix} -0.2042 \\ 0 \end{bmatrix}}_{\text{A}} + \underbrace{\begin{bmatrix} 0.2042 \\ -0.3451 \end{bmatrix}}_{\text{extra joint velocities}} = \underbrace{\begin{bmatrix} 0 \\ -0.3451 \end{bmatrix}}_{\text{B}} \ \text{rad/s}.$$

A and B give the same horizontal tip velocity. The extra motion therefore contributes **zero horizontal velocity**.

Joint velocities with zero effect on the chosen task form its **null space**:

$$\mathbf{J}_x \, \dot{\mathbf{q}}_{\text{null}} = 0.$$

Vertical velocity may still change.




The blue arrows show the same extra joint velocities: adding them takes A to B.

code/two_link.py · code/make_figures.py · essentials §7

# The arm can change posture while the gripper stays fixed

The gripper is at the required grasp pose. Keep its **position and orientation**, while changing the arm's posture.

Several joints can move together so their effects at the gripper **cancel**.

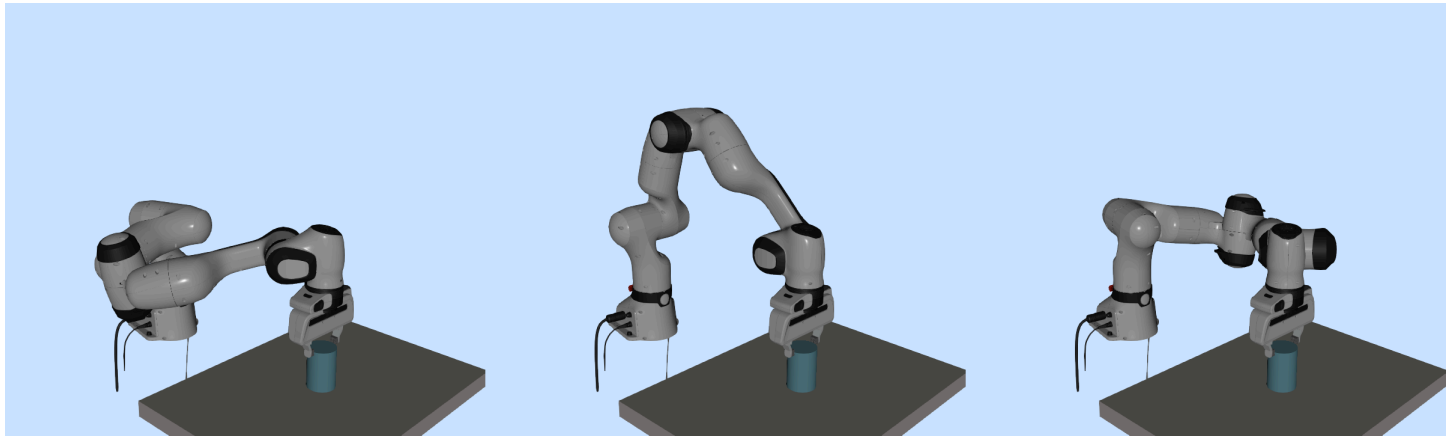
Now use the **full-pose geometric Jacobian**:

$$\mathbf{J}\,\dot{\mathbf{q}}_{\text{null}} = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v} \end{bmatrix} = \mathbf{0}.$$

| Panda at a regular configuration | Count |
|---|---|
| Joint angular velocities | $7$ |
| Independent gripper-motion components held at zero | $6$ |
| Independent coordinated motion patterns remaining | $1$ |

That one remaining pattern moves several joints together while the gripper does not move. The next demo shows it: the elbow swings and the gripper pose stays fixed.

# Live demo: redundancy, the elbow swings, the hand stays



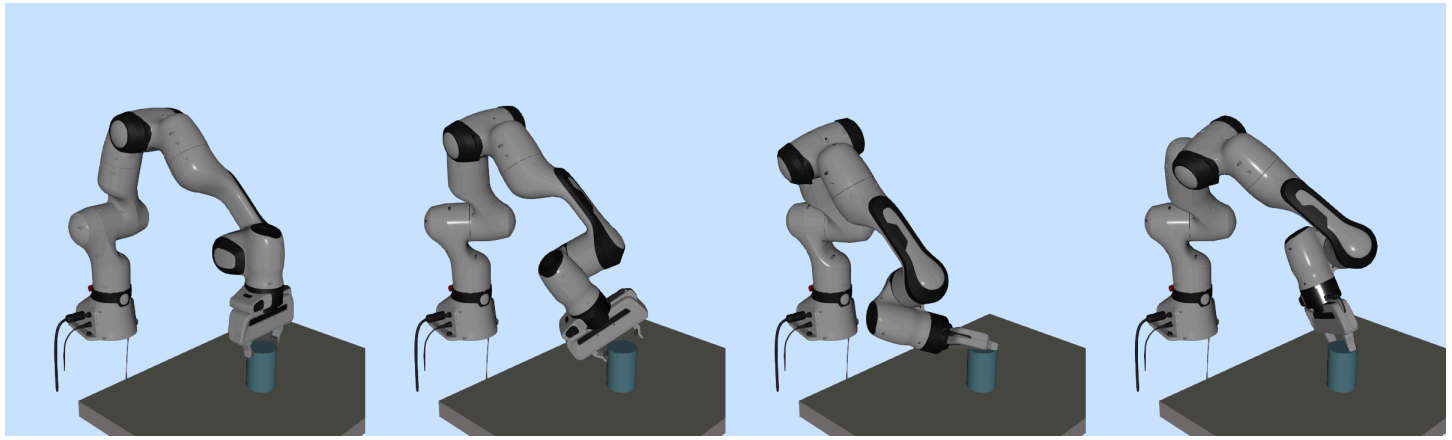
The flange pose stays fixed. The tool axis points down while the elbow rises and falls.

Watch the joints move **together**. The arm changes posture while the gripper keeps its position and orientation.

The demo repeats small null-space moves and corrects drift; the right panel's seven readouts show the joint angles in degrees.

drake/01b-redundancy-self-motion ▶ run it live

# Same demo, only the fingertip position fixed



Now hold the tool-centre point between the fingers. The gripper may turn while this point stays in place.

Relaxing the orientation requirement allows **more coordinated joint motions**.

At full position-task rank, seven joints and three position components leave **four independent motion patterns**.

The held point is now the tool center, $10.3$ cm beyond the flange.

drake/01b-redundancy-self-motion/tip_only_panda.py --app ▶ run it live

# A velocity becomes a small position command

**Back to the two-link arm** of slide 36 and its upward tip command. The Panda demos showed what redundancy leaves free; these two slides turn a joint velocity, and then a target, into a joint change.

If the robot accepts joint positions, integrate over one short control period:

$$\mathbf{q}_{\text{next}} = \mathbf{q} + \Delta t\, \dot{\mathbf{q}}$$

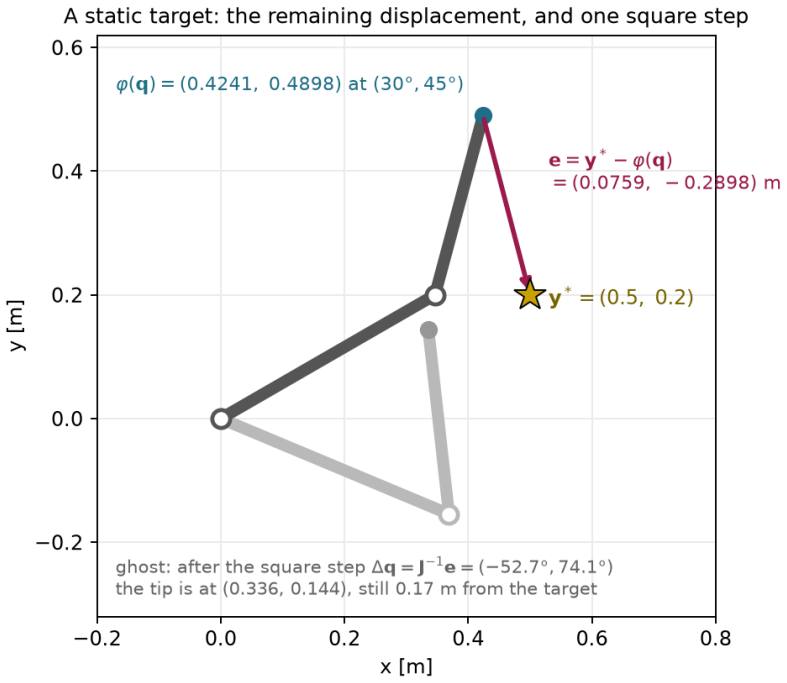
Upward tip command: $\dot{\mathbf{x}} = (0, 0.1)$ m/s.

At our pose $\dot{\mathbf{q}} = (0.3415, -0.5772)$ rad/s; with $\Delta t = 0.01$ s:

$$\Delta\mathbf{q} = (0.003415, -0.005772)\ \text{rad}$$

About **1 mm upward**. Measure the new configuration and recompute.

# From a target back to a joint change




| quantity | value |
|---|---|
| target $\mathbf{y}^*$ | $(0.5, 0.2)$ m |
| tip $\varphi(\mathbf{q})$ at $(30^\circ, 45^\circ)$ | $(0.4241, 0.4898)$ m |
| **displacement** $\mathbf{e} = \mathbf{y}^* - \varphi(\mathbf{q})$, no time step | $(0.0759, -0.2898)$ m |
| local model | $\mathbf{J}(\mathbf{q})\,\Delta\mathbf{q} \approx \mathbf{e}$ |
| $\det \mathbf{J}$ | $0.0849$: square, invertible |
| step $\Delta\mathbf{q} = \mathbf{J}^{-1}\mathbf{e}$ | $(-52.7^\circ, 74.1^\circ)$, far past where the first-order model held |

- **Own now:** the local problem and this one step
- **Deck 04 derives:** how far to trust it, the step for any $\mathbf{J}$, and "closest" when no $\mathbf{q}$ reaches $\mathbf{y}^*$

code/two_link.py (k) · essentials §7; Appendix C, Figure C.1

# What an action asks the robot-side stack to do

| Policy output | Typical conversion | Still robot-dependent |
|---|---|---|
| Joint targets or deltas | Track the targets; no IK conversion of this action | Joint meaning, limits, safety checks |
| End-effector pose | Form geometric error; solve IK | Reachability, posture, collisions |
| End-effector velocity | Differential IK; integrate if needed | Jacobian, rates and limits |

**“Delta” needs a frame, units and an update rule.**

A common task-space interface can help transfer; it does not guarantee it.

What does learning replace, keep, and assume?

# A1 is out; next: optimization

- **A1, the classical stack** (due Oct XX)

- **Deck 04 (3.1–3.2):** gradient descent, Newton, Gauss-Newton; the bridge slide's step derived, damped and repeated, on this arm
- **Backup:** motion gains, extra singularity examples and demo controls; algebra checks; damping, numerical IK and the null-space projection

# Backup slides

Determinant algebra, inverse checks, the failure-mode recap, damping and numerical IK (both derived in Deck 04), and the optional null-space construction.

# $\det \mathbf{J} = l_1 l_2 \sin q_2$

$$\det \mathbf{J} = l_1 l_2 \sin q_2, \qquad \text{at } (30^\circ, 45^\circ): \; 0.4 \cdot 0.3 \cdot 0.7071 = 0.0849$$

- Two lines of algebra from the two-link Jacobian derivation (notes §4.7); the $l_2^2$ terms cancel
- $q_1$ **is absent**: the base joint decides *where* the tip is, not *how well* it moves
- $\det \mathbf{J} = 0$ exactly when $\sin q_2 = 0$: arm stretched or folded (see the singular-configurations slide)

# Backup: the inverse and a finite-difference check

For a square, nonsingular $2 \times 2$ Jacobian:

$$\mathbf{J}^{-1} = \frac{1}{\det \mathbf{J}} \begin{bmatrix} J_{22} & -J_{12} \\ -J_{21} & J_{11} \end{bmatrix}$$

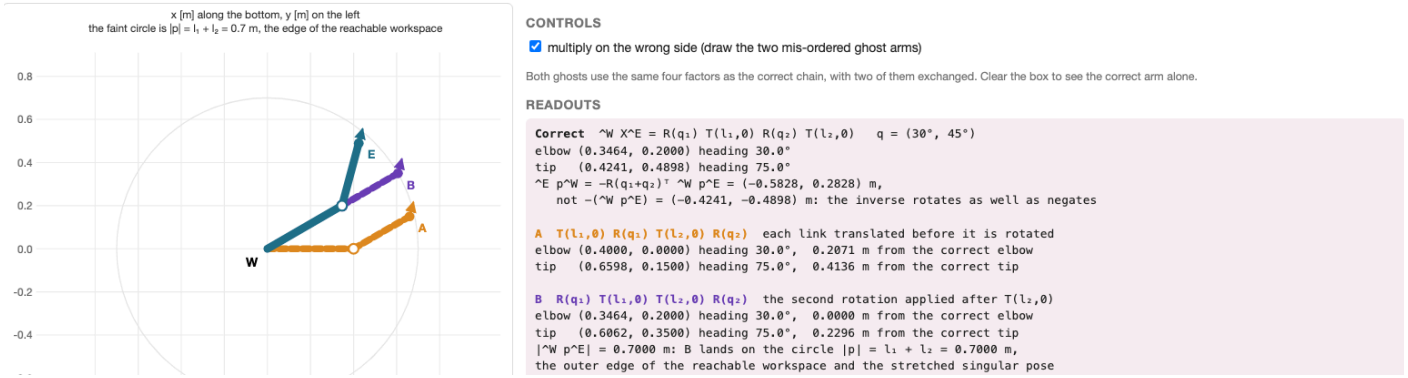
Check a column without differentiating by hand:

$$\mathbf{J}_{:,j} \approx \frac{\varphi(\mathbf{q} + h\mathbf{e}_j) - \varphi(\mathbf{q} - h\mathbf{e}_j)}{2h}$$

$\mathbf{e}_j$ selects joint $j$; $h = 10^{-6}$ rad gives agreement to $3.7 \times 10^{-11}$ here.

# Failure modes of the day

- **Order of multiplication**: right acts in the body frame, left in the world frame
- **The missing** $1$: $(\mathbf{p}, 0)$ rotates but does not translate
- **Rows versus columns**: $\mathbf{J}$ is $m \times n$; $\mathbf{J}\dot{\mathbf{q}}$ needs an $n$-vector
- **"Singular means stuck"**; and trusting $\det \mathbf{J}$ or the rank alone instead of $\sigma_{\min}$




modules/two-link-arm, wrong-order section: ghost A elbow 21 cm off and tip 41 cm off; ghost B elbow right and tip 23 cm off; heading right every time

# Damping: accept error instead of excessive joint speed

A weak motion direction needs large joint rates to produce a small tip velocity.

Damped least squares balances two costs:

$$\min_{\dot{\mathbf{q}}} \underbrace{\|\mathbf{J}\dot{\mathbf{q}} - \dot{\mathbf{x}}\|^2}_{\text{tracking error}} + \rho \underbrace{\|\dot{\mathbf{q}}\|^2}_{\text{joint motion}}, \quad \rho > 0$$

Larger $\rho$: less joint motion, more accepted tracking error.

**Damping regularises. Explicit constraints enforce speed, joint and collision limits.**

# Numerical IK: improve a guess, then linearise again

For a **position target**, the remaining displacement is
$\mathbf{e} = \mathbf{y}^* - \varphi(\mathbf{q})$.

$$\Delta\mathbf{q} = \mathbf{J}^{+}(\mathbf{q})\mathbf{e}, \qquad \mathbf{q} \leftarrow \mathbf{q} + \Delta\mathbf{q}$$

**Recompute** the error and Jacobian; repeat. This is the Gauss–Newton step.

- A large step may need shortening or damping
- Unreachable targets and poor initial guesses can prevent convergence
- Full pose: a geometric rotation and position error in the Jacobian's frame; its residual derivative depends on the error convention; this block stays with position

The optimization block derives the step and adds constraints.

# Backup: how to compute a null-space motion

Start with preferred joint velocities $\mathbf{z}$. They would cause task velocity $\mathbf{J}\mathbf{z}$.

The pseudo-inverse finds the smallest joint velocities that produce that same task velocity. Subtract them:

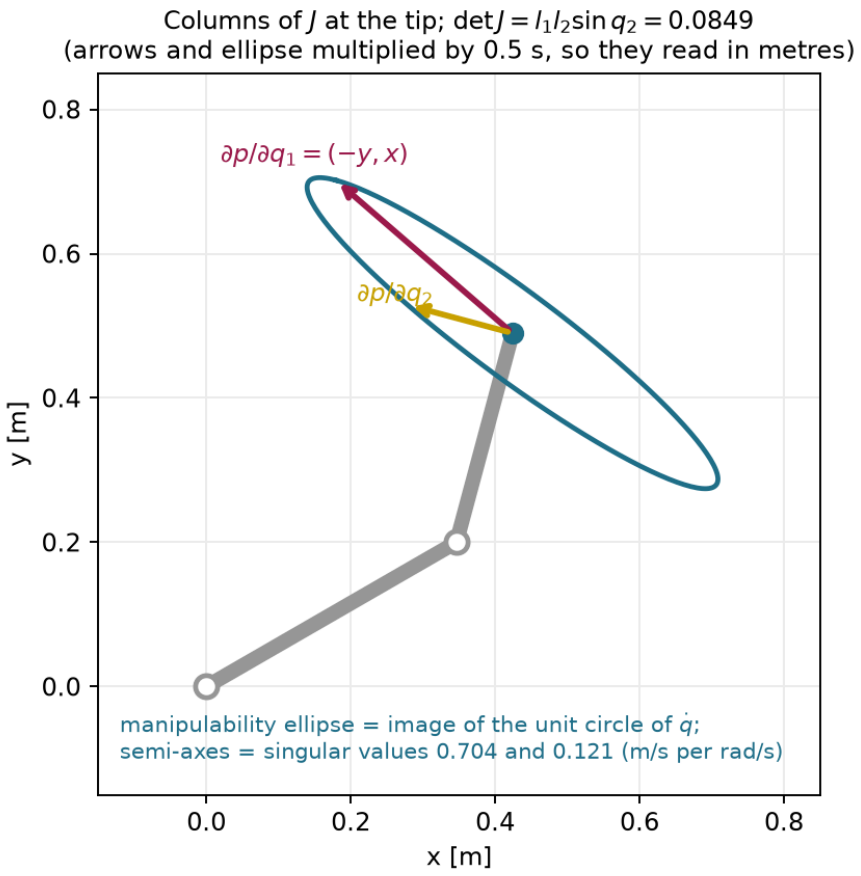
$$\dot{\mathbf{q}}_{\text{null}} = \mathbf{z} - \mathbf{J}^{+}\mathbf{J}\mathbf{z}, \qquad \mathbf{J}\dot{\mathbf{q}}_{\text{null}} = \mathbf{0}.$$

Add this zero-task motion to a primary velocity solution:

$$\dot{\mathbf{q}} = \mathbf{J}^{+}\dot{\mathbf{x}} + \dot{\mathbf{q}}_{\text{null}}.$$

There are $n - \operatorname{rank}\mathbf{J}$ independent null-space directions. This construction uses the plain pseudo-inverse at the current configuration.

# Backup: motion gains, rank and singular values




We return to the **two-link position task** and its $2 \times 2$ **Jacobian** $\mathbf{J}_{\text{pos}}$.

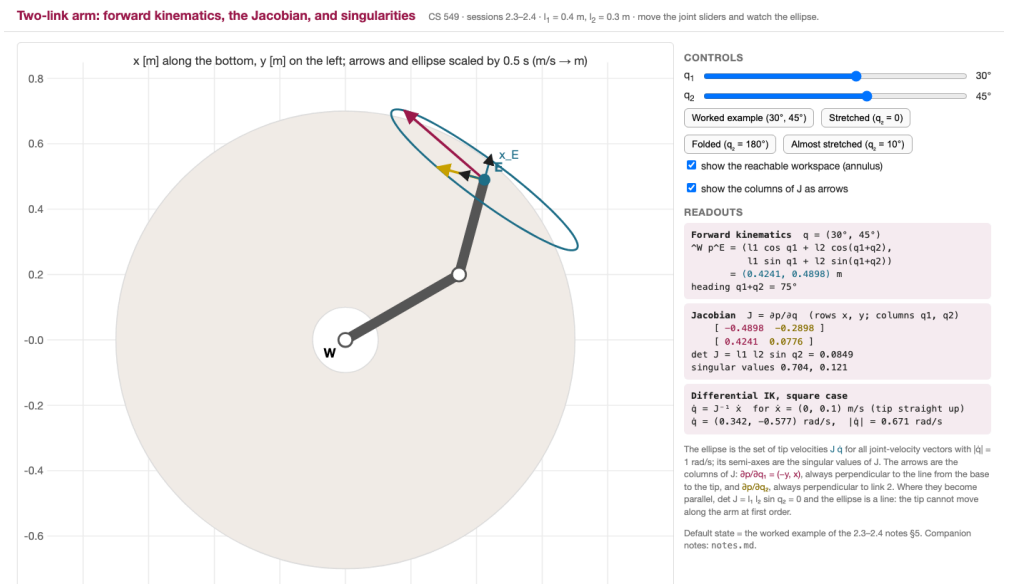
**Rank of** $\mathbf{J}_{\text{pos}}$**:** number of independent tip-velocity directions the joints can produce. Here its rank is **2**.

The **ellipse boundary** traces tip velocities at unit joint-rate norm.

Its radii are the **singular values**: largest and smallest motion gains.

Here: $\sigma_1 = 0.7037$, $\sigma_2 = \sigma_{\min} = 0.1206$ m/rad. The short axis points almost along the arm.

# Backup: explore the two-link module in more detail




modules/two-link-arm/index.html

- Start at the worked example
- $q_2 \to 0$: ellipse collapses, column arrows turn parallel
- Change $q_1$: ellipse turns, shape unchanged
- Predict: upward $0.1$ m/s at $q_2 = 1^\circ$ costs $2\times$, $10\times$ or $30\times$ the joint speed at $45^\circ$? Then read $\dot{\mathbf{q}}$ at $45^\circ, 10^\circ, 1^\circ$
- Press **Stretched**: rank drops from **2 to 1**. The tip can still move sideways.

# Backup: the Panda's zero configuration is singular

Each turning joint adds angular velocity **along its own axis**.

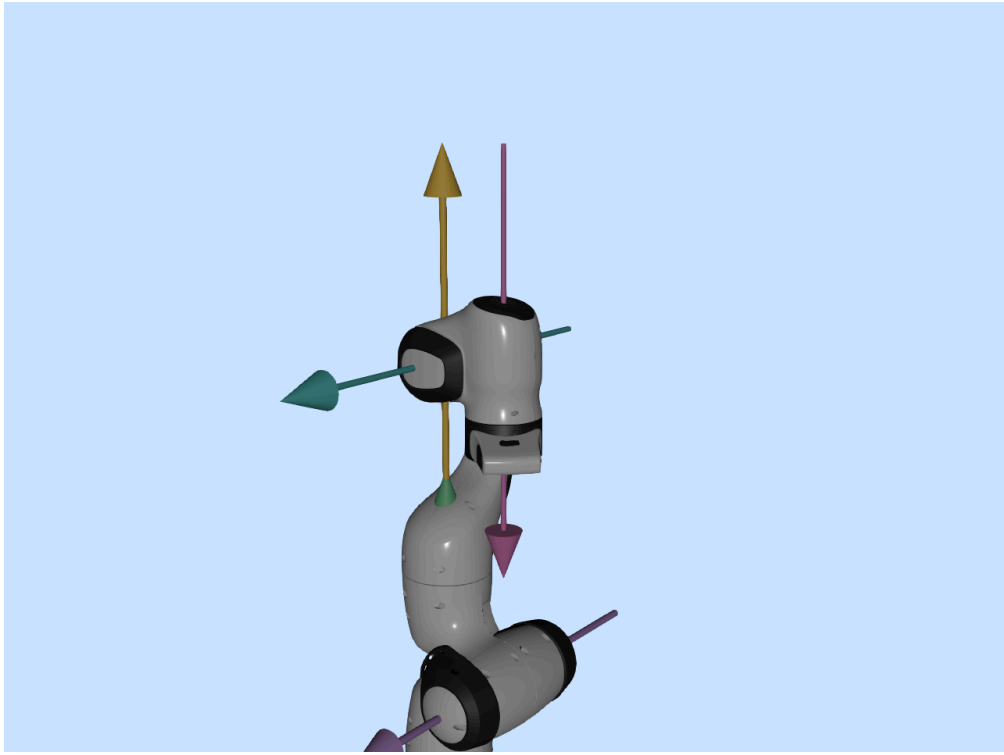
- Joints **1, 3, 5, 7** are parallel to world $z$.
- Joints **2, 4, 6** are parallel to world $y$.

Together, they cannot turn the flange **about world** $x$.

The full $6 \times 7$ geometric Jacobian loses one independent motion:

$$\operatorname{rank}(\mathbf{J}) = 5, \qquad \sigma_{\min} = 0.$$

The zero configuration is outside the Panda model's joint limits.



The arrows show the positive joint-axis directions near the wrist. Open the demo

# Backup: large joint velocities near a singularity

Keep the **two-link position task**, world-up request and shoulder angle fixed:

$$\dot{\mathbf{p}}_{\text{cmd}} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} \text{ m/s}, \qquad q_1 = 30^\circ.$$

| $q_2$ | $\det \mathbf{J}_{\text{pos}}$ | $\dot{\mathbf{q}}$ (rad/s) | $\|\dot{\mathbf{q}}\|$ (rad/s) |
|---|---|---|---|
| $45^\circ$ | $0.0849$ | $(0.3415, -0.5772)$ | $0.67$ |
| $10^\circ$ | $0.0208$ | $(0.9254, -1.8852)$ | $2.10$ |
| $1^\circ$ | $0.0021$ | $(7.3778, -16.9275)$ | $18.47$ |

This request has a component in the arm's weakening motion direction, so the required joint speed grows as the arm straightens.

Open the worked state. Set the elbow to **45°, 10°, 1°** and read the differential-IK joint velocities.